EE1007
Introduction to Computer Science Laboratory

Lab 4
Control statement II

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Lab 4.1
Annual Interest Calculator (30%)

- Test target: std:: **setw**(), std:: **setprecision**(), std::**fixed**(), **for**.
- You are asked to write a program to calculate the **regular saving plan** at the end of each year over a period of 10 years.

```
Enter the annual investment amount: 1000
Enter the annual interest rate (as a decimal, e.g., 0.05 for 5%): 0.05
Year    Amount on deposit
   1             1050.00
   2             2152.50
   3             3310.12
   4             4525.63
   5             5801.91
   6             7142.01
   7             8549.11
   8            10026.56
   9            11577.89
  10            13206.79
```

Lab 4.1
Annual Interest Calculator (30%)

```
for (int year = 1; year <= 10; year++) {
    totalAmount *= (1.0 + annualRate);          // Apply annual interest
    std::cout << std::setw(4) << year << std::setw(21) << std::fixed << std
}
```



```
for (int year = 1; year <= 10; year++) {
    totalAmount += annualInvestment;            // Add the annual investment
    totalAmount *= (1.0 + annualRate);          // Apply annual interest
    std::cout << std::setw(4) << year << std::setw(21) << std::fixed << std::se
}
```


Lab 4.2
Company Investment Comparison (30%)

- Test target: while, continue, break.
- You are asked to compare the performance of three companies based on their annual interest rates.

```
Enter the annual interest rate for Andy company: 0.05
Enter the annual interest rate for NCU company: 0.04
Enter the annual interest rate for EE company: 0.03

1. Andy company (5.00%)
2. NCU company (4.00%)
3. EE company (3.00%)
4. Exit this program
Please select a company (1-3) or enter 4 to exit: 1
Enter the monthly investment amount: 1000
Year    Amount on deposit
   1             12600.00
   2             25830.00
   3             39721.50
   4             54307.58
   5             69622.95
   6             85704.10
   7            102589.31
   8            120318.77
   9            138934.71
  10            158481.45
```

Lab 4.2
Company Investment Comparison (30%)

```cpp
// Prompt the user for input of interest rates with validation
while (true) {
    std::cout << "Enter the annual interest rate for Andy company: ";
    std::cin >> rateAndy;
    if (rateAndy < 0.00 || rateAndy > 1.00) {
        std::cout << "Out of range, interest rate should be in the range [0.00, 1.00]\n";
        continue;
    }
    break;
}
```

```cpp
for (int year = 1; year <= 10; ++year) {
    totalAmount += monthlyInvestment * 12;
    totalAmount *= (1.0 + annualRate);
    std::cout << std::setw(4) << year << std::setw(21) << totalAmount << std::endl;
}
```


Lab 4.3
Luggage Allocation (30%)

- Test target: observation.
- Write a program to determine if a given total weight can be distributed **exactly** using a limited number of boxes with different capacities.

Lab 4.3
Luggage Allocation (30%)

- Brute Force Search
  - Simple but time wasted

```
for (int i = 0; i <= box1kg; ++i) {
    for (int j = 0; j <= box5kg; ++j) {
        for (int k = 0; k <= box10kg; ++k) {
            for (int l = 0; l <= box25kg; ++l) {
                for (int m = 0; m <= box50kg; ++m) {
                    int total = i * 1 + j * 5 + k * 10 + l * 25 + m * 50;
                    if (total == total_weight) {
                        can_distribute_exactly = true;
                        break;
                    }
                }
            }
        }
    }
}
```


Lab 4.3
Luggage Allocation (30%)

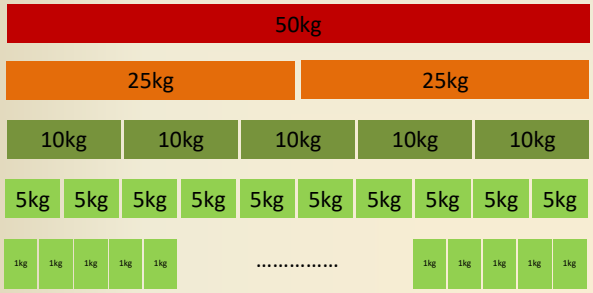
- Brute Force Search
  - Simple but time wasted

✕ 12 Time Exceeded

#	Status	Time Cost	Memory Cost
#1	✓ Accepted	1ms	284 KiB
#2	✓ Accepted	1ms	276 KiB
#3	✕ Time Exceeded	≥1001ms	≥276 KiB
#4	✕ Time Exceeded	≥1001ms	≥352 KiB
#5	✕ Time Exceeded	≥1101ms	≥276 KiB

Lab 4.3
Luggage Allocation (30%)

- **Prioritizing larger boxes**: Maximize weight reduction quickly.



Lab 4.4
Banking System (20%)

- Test target: **switch**, **systematic thinking**.
- Create a simple banking system that allows a user to perform basic operations.

Function	Description
Deposit	Increase balance by a specified amount.
Withdraw	Decrease balance by a specified amount, if funds are sufficient.
Borrow	Add the loan amount to the balance, with 5% annual interest applied upon repayment.
Repay	Repay borrowed amount with interest based on the loan duration.
Exit	Terminate the program.

Lab 4.4
Banking System (20%)

◆ Switch

```
if (choice == EXIT) {
    //...
    return 0;
}
```

```
double amount;
switch (choice) {
    case DEPOSIT: {  // Deposit
        //...
        break;
    }
    case WITHDRAW: {  // Withdraw
        //...
        break;
    }
    case BORROW: {  // Borrow
        //...
        break;
    }
    case REPAY: { // Repay
        //...
        break;
    }
    default: {
        //...
        break;
    }
}
```


Lab 4.4
Banking System (20%)

◆ Repay

```
case REPAY: {
    std::cout << "Current balance: $" << balance << "\n";
    if (loanAmount == 0) {
        std::cout << "No outstanding loan to repay.\n\n";
        break;
    }
    int years;
    std::cout << "Enter the number of years since the loan: ";
    std::cin >> years;
    double totalRepay = loanAmount + (loanAmount * interestRate * years);

    std::cout << "Total amount to repay: $" << totalRepay << "\n";

    if (balance >= totalRepay) {
        balance -= totalRepay;
        loanAmount = 0;  // Reset loan amount after full repayment
        std::cout << "Repayment successful.\n";
        std::cout << "Balance after repayment: $" << balance << "\n\n";
    } else {
        std::cout << "Insufficient balance to repay the loan.\n\n";
    }
    break;
}
```

Q&A

◆ Time Exceeded

✕ 12 Time Exceeded

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